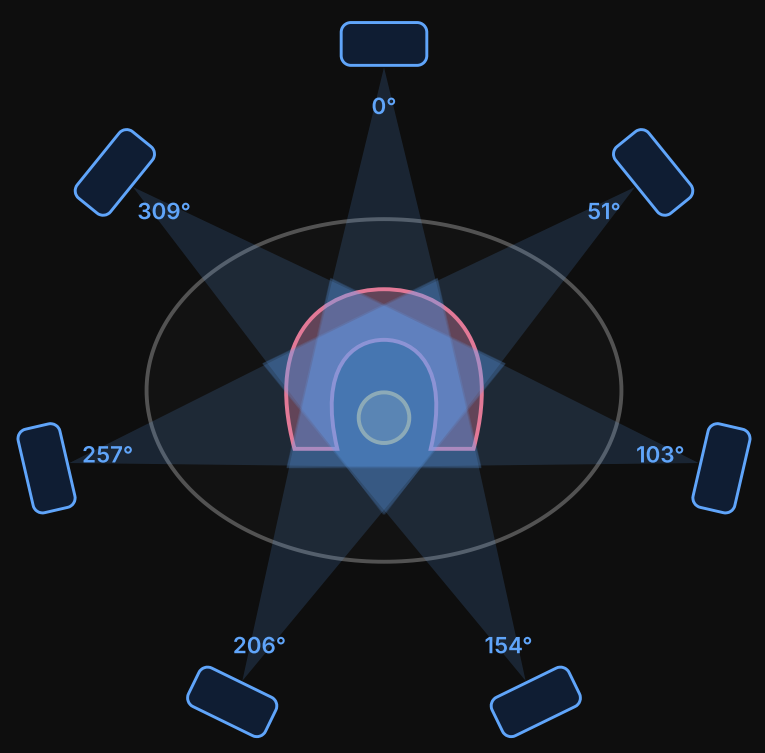
RADIOTHERAPY DELIVERY TECHNIQUES

IMRT vs VMAT: same goal, different motion

Both modulate the beam to wrap the dose around the tumour and spare healthy organs. The difference is how the linac moves.

IMRT

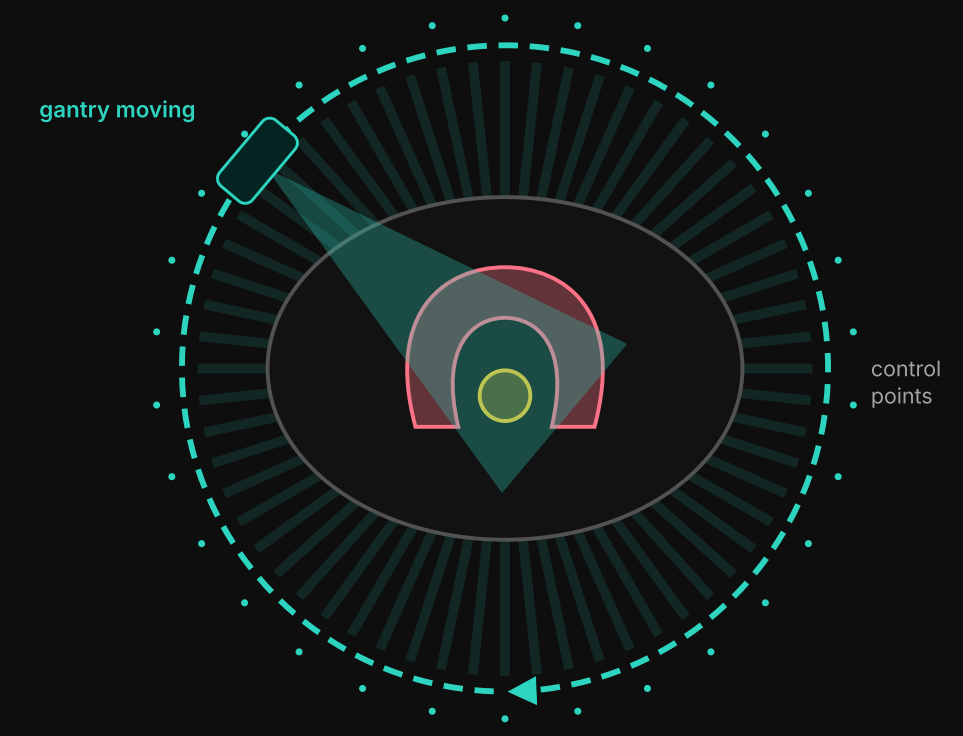
Fixed beams 5–9 gantry angles



gantry **stops** at each angle, then the beam turns on

VMAT

Rotating arc 1–2 arcs around the patient



beam stays **on** while the gantry rotates around the patient

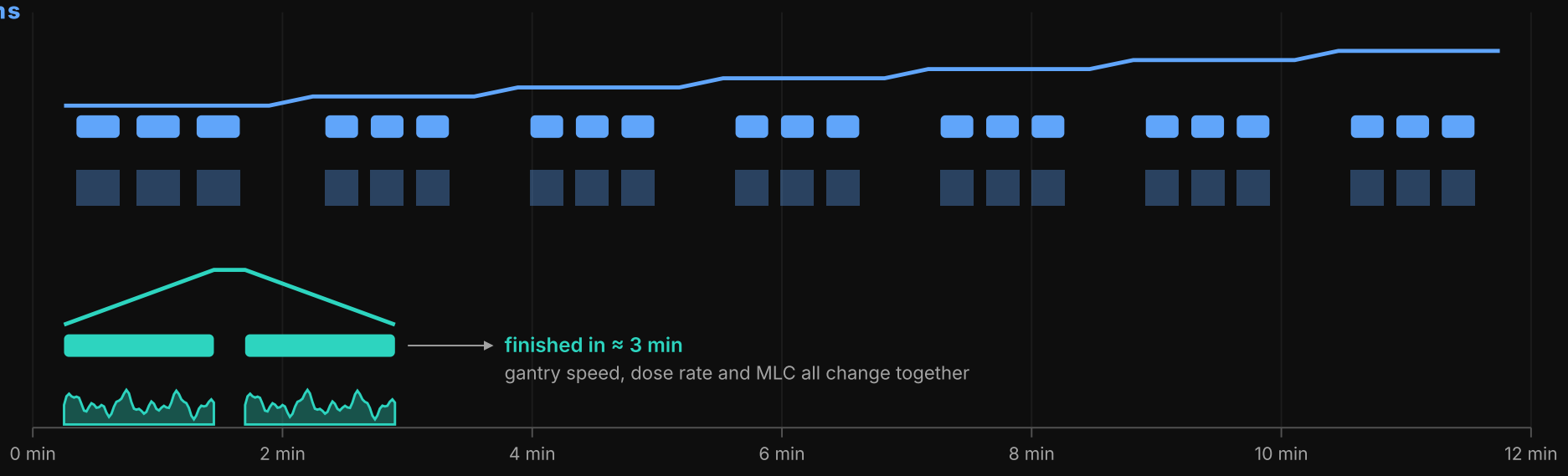
target (tumour) wrapping around

organ at risk (e.g. spinal cord)

Both techniques shape the dose around the organ at risk. They differ in **how the machine moves**.

DURING DELIVERY

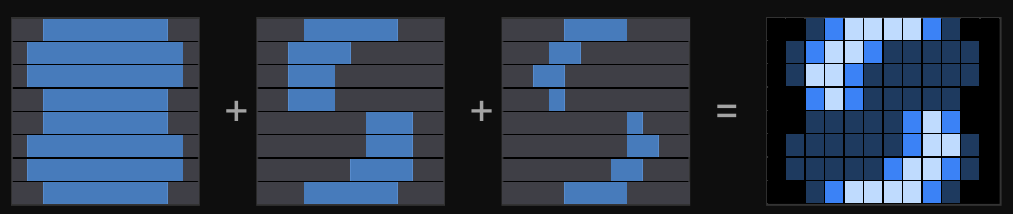
What the machine is doing over time typical, illustrative timings



IMRT

Intensity built at each angle

At one gantry angle: stack several MLC shapes

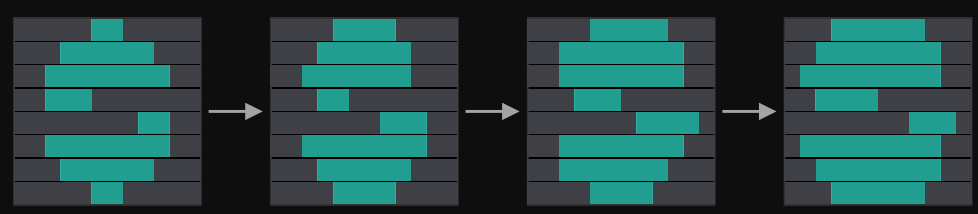


Step-and-shoot: beam off between shapes. Sliding window: leaves sweep, beam on.

VMAT

Intensity built across angles

Along the arc: one MLC shape per control point



The shape morphs a little every ≈ 2° (≈ 180 control points per full arc) as the gantry turns.

Treatment time

IMRT ≈ 10–15 min

VMAT ≈ 2–4 min

Faster delivery means less time for the patient to move.

Monitor units

IMRT more MU

VMAT often fewer

Fewer MU means less leakage dose to the rest of the body.

Low-dose spread

IMRT fewer entry paths

VMAT all angles

VMAT spreads a low dose over a larger volume of tissue.

Machine QA demands

IMRT MLC, output

VMAT + gantry speed, dose rate

VMAT also needs dynamic tests of everything moving together.

Same calibration underneath. Both techniques deliver monitor units that were calibrated by TRS-398 in a static 10×10 cm² field. An error in the reference calibration shifts every IMRT and VMAT plan equally. Both also need patient-specific QA before treatment.

Other arc names. IMAT was the early multi-arc form of arc IMRT. **RapidArc** (Varian) and **VMAT/SmartArc** (Elekta, Philips) are vendor names for VMAT. **Tomotherapy** delivers IMRT in a helical arc with a binary MLC.